

November 25, 2025

1. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is  $25\text{ ns}$ ; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is  $7.5\text{ ns}$ . What is the propagation velocity of the particle beam?

The Lorentz factor is

$$\gamma = \frac{\Delta t}{\Delta \tau} = \frac{25}{7.5} = \frac{10}{3} \approx 3.3333333.$$

Using  $\gamma = 1/\sqrt{1 - v^2/c^2}$  we obtain

$$v = c\sqrt{1 - \frac{1}{\gamma^2}} = c\sqrt{1 - \left(\frac{3}{10}\right)^2} \approx 0.953939\text{ }c.$$

Thus the beam propagates at approximately 0.95394 times the speed of light.

2. A certain bacterium doubles its population in 25 days. Two such bacteria are placed in a spaceship and sent on an 800-day space journey. During the trip, the spaceship moves at a constant speed of  $0.995\text{ }c$ . How many bacteria will be on board at the moment of return?

The proper time experienced on board is shorter than the laboratory time. The ship frame time is

$$\Delta \tau = \Delta t \sqrt{1 - v^2/c^2} = 800 \sqrt{1 - (0.995)^2} \text{ days} \approx 79.89994 \text{ days}.$$

Exponential growth with doubling time 25 days gives

$$N = N_0 2^{\Delta \tau/25} = 2 \cdot 2^{79.89994/25} \approx 18.33.$$

Approximately 18 bacteria will be on board at return (the continuous model gives about 18.3 individuals).

3. A particle moves with a velocity such that  $\frac{v}{c} = 0.9999$ . Determine the value of  $\frac{m}{m_0}$ .

The relativistic mass factor equals the Lorentz factor

$$\frac{m}{m_0} = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - (0.9999)^2}} \approx 70.712446.$$

4. Determine the relativistic mass and the velocity of an electron whose kinetic energy is 1 MeV ( $1.6 \cdot 10^{-13}\text{ J}$ ). The rest mass of the electron is  $9.11 \cdot 10^{-31}\text{ kg}$ .

From the relation between kinetic energy and gamma factor,

$$E_{kin} = (\gamma - 1)m_0c^2,$$

we obtain

$$\gamma = 1 + \frac{E_{kin}}{m_0c^2} \approx 1 + \frac{1.0 \cdot 10^6 \text{ eV} \cdot 1.6 \cdot 10^{-19} \text{ J/eV}}{(9.11 \cdot 10^{-31} \text{ kg})(299792458 \text{ m/s})^2} \approx 2.9541604.$$

The relativistic mass is

$$m = \gamma m_0 \approx 2.9541604 \cdot 9.11 \cdot 10^{-31} \text{ kg} \approx 2.69124 \cdot 10^{-30} \text{ kg}.$$

The speed follows from

$$v = c \sqrt{1 - \frac{1}{\gamma^2}} \approx 0.94096436 c \approx 2.82094 \cdot 10^8 \text{ m/s}.$$

5. Calculate the rest energy of a proton, corresponding to its rest mass of  $1.67 \cdot 10^{-27} \text{ kg}$ , in joules and in electronvolts. ( $1 \text{ MeV} = 1.6 \cdot 10^{-13} \text{ J}$ )

The rest energy is

$$E = mc^2 = (1.67 \cdot 10^{-27})(299792458)^2 \approx 1.50092115 \cdot 10^{-10} \text{ J}.$$

In mega-electronvolts,

$$E \approx \frac{1.50092115 \cdot 10^{-10}}{1.6 \cdot 10^{-13}} \text{ MeV} \approx 938.08 \text{ MeV}.$$

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